Gihwan Kim

gihwankim1214@gmail.com

Work Experience

Applied Scientist May. 2026 - Present
AWS Center for Quantum Computing

Research Scientist Intern Sep. 2024 - Dec. 2024
AWS Center for Quantum Computing
- Theoretical and numerical modeling towards implementation of simultaneous long-range two-qubit gates with dual-rail transmons.

Research Scientist Intern Aug. 2023 - Nov. 2023
AWS Center for Quantum Computing
- Computer-aided design and simulation of superconducting circuits and qubits.

Education

California Institute of Technology Aug. 2019 - Jun. 2026
Ph.D. in Applied Physics
M.S. in Applied Physics

Seoul National University Mar. 2015 - Feb. 2019
B.S. in Electrical and Computer Engineering

Research Interests

Circuit QED, Waveguide QED, Quantum Information, Quantum Computation

Research Experience

Quantum Photonics Group Sep. 2019 - Present
Advisor: Prof. Oskar Painter, California Institute of Technology

- Experimental study of quantum information science with superconducting circuits and qubits.
- Proposing and analyzing novel architectures and building blocks for superconducting quantum processors.
- Design, electromagnetic simulation, and fabrication of superconducting circuits.
- Cryogenic microwave circuit engineering and operation of ^{3}He/^{4}He dilution refrigerators.
- Development of control software and databases for qubit control.

Publications

1. J. S. C. Hung et al. (**G. Kim** among coauthors), Fast, high-Fidelity erasure detection of dual-rail qubits with symmetrically coupled readout, *under review*, arXiv:2604.16292 (2026)
2. **G. Kim** et al., A tunable, modeless, and hybridization-free cross-Kerr coupler for miniaturized superconducting qubits, *under review*, arXiv:2602.03186 (2026)
3. A. V. Gorshkov et al. (**G. Kim** among coauthors), Cavity-mediated cross-cross-resonance gate, *under review*, arXiv:2506.03239 (2025)
4. **G. Kim***, A. Butler* et al., Fast unconditional reset and leakage reduction of a tunable superconducting qubit via an engineered dissipative bath, *Phys. Rev. Appl.* 24, 014013 (2025)
5. V.S. Ferreira*, **G. Kim***, A. Butler* et al., Deterministic generation of multidimensional photonic cluster states with a single quantum emitter, *Nat. Phys.* 20, 865870 (2024)

CONFERENCES

APS Global Physics Summit 2026, Speaker Mar. 2026
A tunable, modeless, and hybridization-free cross-Kerr coupler for a miniaturized quantum processor architecture

Les Houches Workshop in Fault-tolerant quantum computing, Speaker Apr. 2025
Simultaneous long-range gates with dual-rail transmons

APS Global Physics Summit 2025 Mar. 2025
Long-range gates in superconducting dual-rail qubits - Part 1, 2

APS March Meeting 2024, Speaker Mar. 2024
Rapid unconditional active reset of frequency-tunable superconducting qubits via a metamaterial waveguide Purcell filter

Gordon Research Conference (Quantum Science), Poster July 2022
Deterministic generation of multidimensional photonic cluster states with a single quantum emitter **APS March Meeting 2022, Speaker of Part 2** Mar. 2022

Deterministic generation of multidimensional microwave photonic cluster states with a single quan-tum emitter - Part 1, 2

PEER REVIEW SERVICES

Reviewer for Physical Review Letters (1 manuscript)
Co-reviewer for Nature Communications (1 manuscript)

SOFTWARE AND EXPERIMENTAL TOOLS

Python, Julia, MATLAB, C/C++, Palace, COMSOL, SONNET, QuTiP, scQubits
Bluefors Dilution Refrigerator, Zurich Instruments HDAWG, Keysight Digtizer
Quantum Machines OPX+, Octave, QDevil, and OPX1000

UNDER-GRADUATE RESEARCH INTERNSHIP

Quantum Information and Quantum Computing Lab Sep. 2018 - June 2019
Advisor: Prof. Taehyun Kim, Seoul National University
- Developed a software for laser frequency regulation using piezo-electric actuator, wave-meter monitoring, and a PID algorithm, aiding in trapping and cooling $^{171}Yb^{+}$ and $^{174}Yb^{+}$ ions.

High Performance Computer System Lab, Bachelor's Thesis Sep. 2017 - June 2018
Advisor: Prof. Jangwoo Kim, Seoul National University
- Proposed and developed a simulator for CMOS devices operating at 77 Kelvin by extrapolating the BSIM model to low temperature, and conducted numerical optimization of an SRAM at this temperature condition.

TEACHING

Teaching Assistant, Seoul National University
Introduction to Quantum Computing and Information Mar. 2019 - June 2019

LEADERSHIP

Treasurer, Caltech Korean Graduate Student Association Jan. 2022 - Dec. 2022